TM Forum Specification

Fraud Management User Guide

TMF770

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Introduction

The following document is the specification of the REST API for Fraud Management. It includes the model definition as well as all available operations. In the telecommunications industry, fraud detection and prevention are critical to protecting both service providers and their customers. As digital transactions increase and new attack vectors emerge, Communication Service Providers (CSPs) need a standardized approach to evaluate fraud risk across multiple dimensions including identity verification, network authentication, and behavioral analysis.

The Fraud Management API provides a standardized mechanism for evaluating fraud risk associated with a telecommunications customer or transaction. The API enables requesting parties (such as merchants, banks, or internal systems) to submit fraud evaluation requests and receive comprehensive risk assessments. The API consists of a task-based resource that coordinates multiple fraud signal checks in a consistent manner. A fraud evaluation is performed based on criteria specified by the requester, covering risk profiling, network authentication, fraud scoring, and profile matching.

A canonical JSON example of `EvaluateFraudRisk` (request and response shape, polymorphic attributes, and nested criteria) is provided in the API documentation package as `resource-samples/EvaluateFraudRisk.json`. The examples in this user guide and the operation/notification samples follow the same schema patterns and TM Forum example rules: `@baseType` appears only for functional inheritance (in this API, typically `ContactMedium` for phone, email, and geographic-address contact mediums, and `Event` for event payloads). It is not set to the same value as `@type`, nor used for technical base classes such as `EntityRef` or `PartyRef`.

The Fraud Management API goal is to provide fraud risk evaluation capabilities, with 1 task resource:

- EvaluateFraudRisk

EvaluateFraudRisk allows to:

- request a fraud risk evaluation for a target identifier (e.g. phone number) against a set of configurable fraud evaluation criteria. The requester provides the target, criteria, and related party information and expects to receive a comprehensive fraud evaluation result.
- in the request, the following attributes could/must be filled:
  - fraudTargetIdentifier, the contact medium (e.g. `PhoneContactMedium` with `phoneNumber`) identifying the subject of the fraud evaluation

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- fraudEvaluationCriteria (`FraudEvaluationCriteria`), specifying which fraud checks to perform:
  - fraudRiskProfileCriteria (SIM tenure, device tenure, line tenure, payment method, call forwarding, service status, SIM swap)
  - fraudNetworkAuthenticationCriteria (call verification)
  - fraudScoreCriteria (whether explanation is required)
  - fraudProfileMatchCriteria: profile data for matching — `email` as `EmailContactMedium` (with `emailAddress`), `partyName` as a string, and `address` as `GeographicAddressContactMedium` (street, city, country, post code, state or province, optional `geographicAddress` reference, etc.)
- relatedParty, parties involved in the evaluation (e.g. merchant, customer), using `RelatedPartyRefOrPartyRoleRef` with `partyOrPartyRole` (e.g. `PartyRef` with `id`, `name`, `@referredType`, optional `href`)
- requestedFraudEvaluationDate, the date/time when the evaluation was requested

- Polymorphic metadata follows TMF guidelines: `@type` on entities and refs as required; `@referredType` on entity references; `@schemaLocation` optional (mandatory when illustrating an extension). `@baseType` only where there is meaningful inheritance — see the note above and Support of polymorphism and extension patterns.
- In the response, the API will provide the fraud evaluation result including:
  - fraudRiskProfileResult (`FraudRiskProfileResult`: SIM/device/line tenure, payment method, call forwarding status, service status, SIM swap status, etc.)
  - fraudNetworkAuthenticationResult (`FraudNetworkAuthenticationResult`: call verification status such as VERIFIED, NOT_VERIFIED, PENDING)
  - fraudScoreResult (`FraudScoreResult`: fraud score as a string with optional explanation)
  - fraudProfileMatchResult (`FraudProfileMatchResult`: `globalMatchScore` and `profileMatch` array of `ProfileMatch` with `matchElement` and `matchScore`)

The API supports both synchronous and asynchronous execution modes:

- POST may return **200** (results without persisting a task) or **201** (results with a persisted task), depending on server policy.
- The user guide example aligns with a persisted task for auditing (**201 Created**).

Fraud Management API could be used:

- By merchants or banks to verify customer identity during transactions

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- By internal fraud detection systems for account takeover prevention
- As part of a broader fraud prevention workflow integrated with other TMF APIs

Fraud Management API performs the following operations on EvaluateFraudRisk:

- Retrieval of an EvaluateFraudRisk or a collection of EvaluateFraudRisk depending on filter criteria
- Creation of an EvaluateFraudRisk (synchronous or asynchronous)
- Notification of events on EvaluateFraudRisk
  - EvaluateFraudRisk creation
  - EvaluateFraudRisk state change

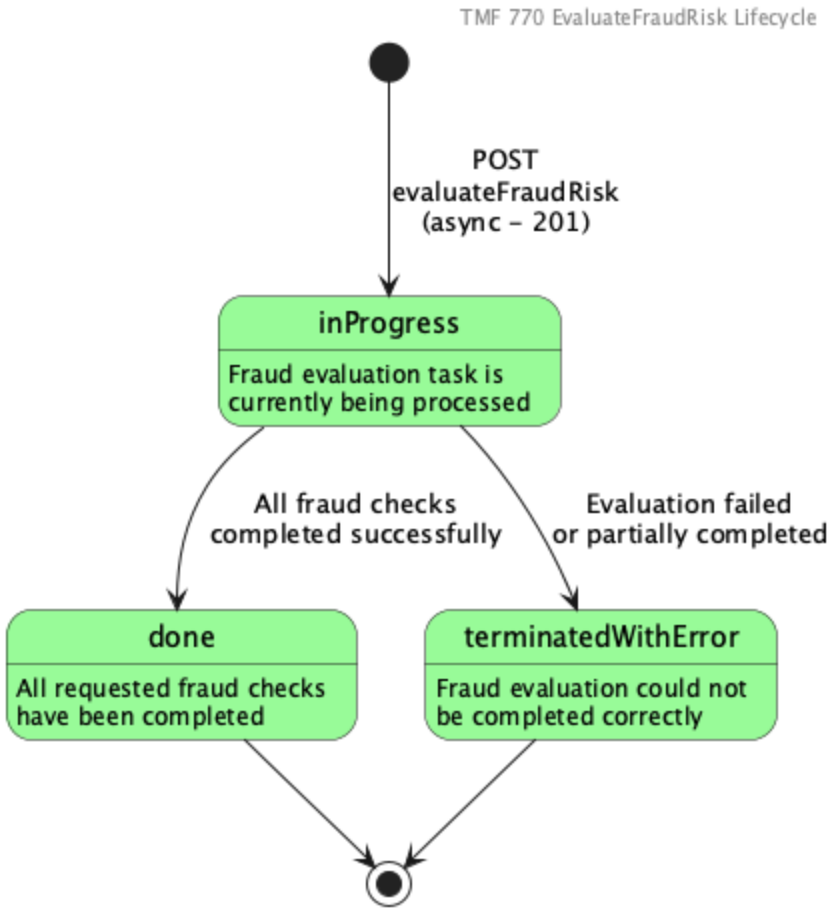
The following Assumptions were considered in the development of this document:

- The Fraud Management system has access to telecom subscriber data (SIM, device, line information)
- The system has access to network authentication services for call verification
- The system has access to fraud scoring engines and profile matching databases

Lifecycle

The state machine specifying the typical state change transitions is provided below. This is standard task based resource lifecycle:

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State	Definition
InProgress	The In Progress state is when the fraud evaluation task is currently being processed. The system is actively performing the requested fraud checks (risk profiling, network authentication, scoring, profile matching).
Done	The fraud evaluation has been completed. All requested fraud checks have been performed and results are available for the requester.
TerminatedWithError	The fraud evaluation task could not be correctly completed. The evaluation is not available or partial due to system errors or unavailable fraud signal sources.

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Sample Use Cases

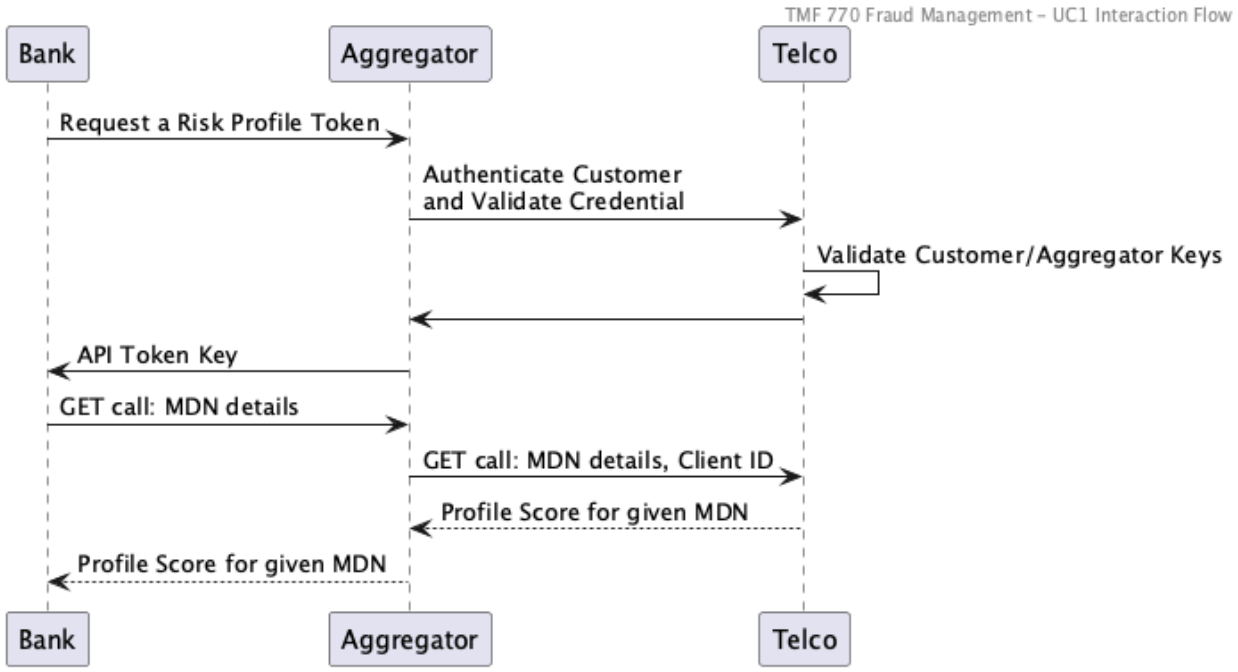
Fraud Management Use Case

The Fraud management API allows the management of the customer fraud profile details.

UC1: A Financial Institution would like to verify if the customer is who he says he is using the Phone number for a given transaction.

A Bank will call this API to validate the customer information. Bank asks and gets their Customer consent to verify his identity using the phone number he has provided to Bank. Bank applications can use this API and pass required and/or optional parameters to the Aggregator application Interface and in turn it passes the details to Telco API with Client ID details. Response can be either True/False or it can be a score based on 0 to 100 with 100 being all validations were successful.

Interaction Flow



Fraud Evaluation Scenario: Suspicious Transaction

Scenario: A merchant reports a suspicious transaction from phone number "555-0123" and requests fraud evaluation.

The request and response below follow the `EvaluateFraudRisk` schema and the same structural patterns as `resource-samples/EvaluateFraudRisk.json`.

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Request Structure


```
{
  "@type": "EvaluateFraudRisk",
  "requestedFraudEvaluationDate": "2025-03-16T10:00:00Z",
  "fraudTargetIdentifier": [
    {
      "@type": "PhoneContactMedium",
      "@baseType": "ContactMedium",
      "phoneNumber": "555-0123"
    }
  ],
  "relatedParty": [
    {
      "@type": "RelatedPartyRefOrPartyRoleRef",
      "partyOrPartyRole": {
        "@type": "PartyRef",
        "id": "merch-001",
        "name": "Example Merchant",
        "@referredType": "Organization"
      },
      "role": "Merchant"
    }
  ],
  "fraudEvaluationCriteria": {
    "@type": "FraudEvaluationCriteria",
    "fraudRiskProfileCriteria": {
      "@type": "FraudRiskProfileCriteria",
      "simTenure": true,
      "deviceTenure": false,
      "lineTenure": true,
      "paymentMethod": false,
      "callForwarding": true,
      "serviceStatus": true,
      "simSwap": true
    },
    "fraudNetworkAuthenticationCriteria": {
      "@type": "FraudNetworkAuthenticationCriteria",
      "callVerification": true
    },
    "fraudScoreCriteria": {
      "@type": "FraudScoreCriteria",
      "isExplanationRequired": true
    },
    "fraudProfileMatchCriteria": {
      "@type": "FraudProfileMatchCriteria",
      "email": {
        "@type": "EmailContactMedium",
        "@baseType": "ContactMedium",
        "emailAddress": "john.doe@gmail.com"
      },
      "partyName": "John Doe",
      "address": {
        "@type": "GeographicAddressContactMedium",
        "@baseType": "ContactMedium",
        "street1": "123 Fake Street",
        "city": "Springfield"
      }
```


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```
    }
  }
}
```

What Happens During Evaluation

1. Direct Checks on the Phone Number (fraudTargetIdentifier):

- How long has this SIM been active?
- Is this number using call forwarding?
- Recent SIM swaps?
- Service status verification

2. Pattern Matching (fraudProfileMatchCriteria):

The API evaluates each provided profile element against fraud databases and watchlists:

- Email "john.doe@gmail.com" matches 85% with known fraudster email pattern
- Address "123 Fake Street" appears in fraud address database
- Name "John Doe" is identified as a generic name often used in fraud attempts

Understanding Match Elements

The `matchElement` property in the response identifies which specific input field was evaluated. Match elements correspond to the `fraudProfileMatchCriteria` inputs:

- `emailAddress` — matches the `email` (`EmailContactMedium`) `emailAddress` value
- `name` — matches the `partyName` string
- `address` — matches the `address` (`GeographicAddressContactMedium`) input
- `phoneNumber` — if included in `fraudProfileMatchCriteria`

Response Example

```
{
  "@type": "EvaluateFraudRisk",
  "id": "eval-suspicious-001",
  "state": "done",
  "fraudEvaluationResult": {
    "@type": "FraudEvaluationResult",
    "fraudRiskProfileResult": {
      "@type": "FraudRiskProfileResult",
      "simTenure": {
        "units": "month",
        "amount": 14
```

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```
      },
      "lineTenure": {
        "units": "month",
        "amount": 24
      },
      "callForwardingStatus": false,
      "serviceStatus": "active",
      "simSwapStatus": false
    },
    "fraudNetworkAuthenticationResult": {
      "@type": "FraudNetworkAuthenticationResult",
      "callVerification": "VERIFIED"
    },
    "fraudScoreResult": {
      "@type": "FraudScoreResult",
      "score": "75",
      "explanation": "Elevated risk from profile matching; SIM and line tenure within
expected range."
    },
    "fraudProfileMatchResult": {
      "@type": "FraudProfileMatchResult",
      "globalMatchScore": "75",
      "profileMatch": [
        {
          "@type": "ProfileMatch",
          "matchElement": "emailAddress",
          "matchScore": "85"
        },
        {
          "@type": "ProfileMatch",
          "matchElement": "name",
          "matchScore": "60"
        },
        {
          "@type": "ProfileMatch",
          "matchElement": "address",
          "matchScore": "80"
        }
      ]
    }
  }
}
```


Fraud Evaluation Criteria

Optional vs Required Fraud Checks

Not all fraud evaluations need all checks. The separate criteria schemas allow:

- **Mix-and-match** - Include only the fraud detection methods you need
- **Conditional logic** - Different fraud scenarios can use different combinations
- **Future expansion** - New fraud detection methods can be added without modifying existing schemas

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Available fraud check types:

- `fraudRiskProfileCriteria`
- `fraudNetworkAuthenticationCriteria`
- `fraudScoreCriteria`
- `fraudProfileMatchCriteria`

Match Score Format

The match score is represented as a string to support multiple scoring formats used by different fraud detection systems:

- **Percentage strings:** "90", "85.5"
- **Decimal confidence scores:** "0.90", "0.85"
- **Categorical ratings:** "HIGH", "MEDIUM", "LOW"
- **Qualified scores:** "90-EXACT", "75-FUZZY"

Example Use Cases:

- High-sensitivity applications (banking)
- Risk management systems
- Hybrid systems combining numerical and qualitative indicators

SIM Swap & Replacement Checks (Post-Issuance)

Scenario: A telecom operator receives a SIM replacement request for phone number "91-98765-43210". Per TRAI's mandated six-step authentication procedure, the operator must verify the request is legitimate before processing. Additionally, the system must enforce the seven-day porting restriction after a SIM swap or replacement.

Regulatory Context:

TRAI has mandated a six-step authentication procedure for processing SIM replacement:

1. Photo Capture by the agent
2. Face match with Aadhaar photo
3. Biometric scan (fingerprint and iris)
4. Aadhaar-based OTP to the registered number
5. Matching old SIM details with KYC documents

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6. Tele-verification through a personalized question set

Additionally, a Unique Porting Code (UPC) cannot be allocated if the request is made within seven days of a SIM swap or replacement, preventing fraudsters from quickly porting away after a fraudulent SIM swap.

Request Structure

```
{
  "@type": "EvaluateFraudRisk",
  "requestedFraudEvaluationDate": "2025-03-16T10:15:00Z",
  "fraudTargetIdentifier": [
    {
      "@type": "PhoneContactMedium",
      "@baseType": "ContactMedium",
      "phoneNumber": "91-98765-43210"
    }
  ],
  "relatedParty": [
    {
      "@type": "RelatedPartyRefOrPartyRoleRef",
      "partyOrPartyRole": {
        "@type": "PartyRef",
        "id": "cust-ind-2025-001",
        "name": "Ramesh Kumar",
        "@referredType": "Individual"
      },
      "role": "Customer"
    },
    {
      "@type": "RelatedPartyRefOrPartyRoleRef",
      "partyOrPartyRole": {
        "@type": "PartyRef",
        "id": "agent-ret-2025-042",
        "name": "Authorized Retailer - Sector 18 Noida",
        "@referredType": "Organization"
      },
      "role": "RetailAgent"
    }
  ],
  "fraudEvaluationCriteria": {
    "@type": "FraudEvaluationCriteria",
    "fraudRiskProfileCriteria": {
      "@type": "FraudRiskProfileCriteria",
      "simTenure": true,
      "simSwap": true,
      "serviceStatus": true,
      "deviceTenure": false,
      "lineTenure": true,
      "paymentMethod": false,
      "callForwarding": true
    },
    "fraudNetworkAuthenticationCriteria": {
      "@type": "FraudNetworkAuthenticationCriteria",
      "callVerification": true
    },
```

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```
    "fraudScoreCriteria": {
      "@type": "FraudScoreCriteria",
      "isExplanationRequired": true
    },
    "fraudProfileMatchCriteria": {
      "@type": "FraudProfileMatchCriteria",
      "partyName": "Ramesh Kumar",
      "address": {
        "@type": "GeographicAddressContactMedium",
        "@baseType": "ContactMedium",
        "city": "Noida",
        "country": "India",
        "postCode": "201301",
        "stateOrProvince": "Uttar Pradesh",
        "street1": "B-42, Sector 18"
      }
    }
  }
}
```

What Happens During Evaluation

1. SIM Swap & Tenure Checks (fraudRiskProfileCriteria):

- Was a SIM swap recently performed on this number?
- How long has the current SIM been active? (A very recent SIM with an immediate replacement request is suspicious)
- Is the line tenure consistent with a genuine long-term customer?
- Is call forwarding enabled? (Fraudsters often enable forwarding to intercept OTPs)
- Is the service currently active?

2. Network Authentication (fraudNetworkAuthenticationCriteria):

- Call verification to the registered number confirms the legitimate subscriber is reachable
- Supports the Aadhaar-based OTP step in TRAI's six-step process

3. Profile Matching (fraudProfileMatchCriteria):

- Verify that the name and address provided at the retail outlet match existing KYC records
- Supports step 5 of TRAI's procedure — matching old SIM details with KYC documents

Response Example

```
{
```

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```
  "@type": "EvaluateFraudRisk",
  "id": "sim-swap-eval-2025-001",
  "state": "done",
  "fraudEvaluationResult": {
    "@type": "FraudEvaluationResult",
    "fraudRiskProfileResult": {
      "@type": "FraudRiskProfileResult",
      "simTenure": {
        "units": "month",
        "amount": 36
      },
      "lineTenure": {
        "units": "month",
        "amount": 60
      },
      "callForwardingStatus": false,
      "serviceStatus": "active",
      "simSwapStatus": false
    },
    "fraudNetworkAuthenticationResult": {
      "@type": "FraudNetworkAuthenticationResult",
      "callVerification": "VERIFIED"
    },
    "fraudScoreResult": {
      "@type": "FraudScoreResult",
      "score": "15",
      "explanation": "Low risk - long SIM tenure (36 months), no recent SIM swap, no
call forwarding, call verified. Seven-day porting restriction not applicable."
    },
    "fraudProfileMatchResult": {
      "@type": "FraudProfileMatchResult",
      "globalMatchScore": "95",
      "profileMatch": [
        {
          "@type": "ProfileMatch",
          "matchElement": "name",
          "matchScore": "98"
        },
        {
          "@type": "ProfileMatch",
          "matchElement": "address",
          "matchScore": "92"
        }
      ]
    }
  }
}
```


Interpretation

A low fraud score (15) with high profile match (95) indicates a legitimate SIM replacement request. The SIM has been active for 36 months with no recent swap history, call forwarding is off, and the subscriber was verified via call. The KYC name and address match closely. The seven-day UPC porting restriction does not apply since no recent SIM swap was detected.

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IMEI & Device Checks

Scenario: A telecom operator needs to validate the IMEI of a device being activated on a new connection. The IMEI must be checked against the CEIR (Central Equipment Identity Register) via the Sanchar Saathi platform to ensure it is not stolen, cloned, or blacklisted.

Regulatory Context:

The Sanchar Saathi platform supports IMEI validation, allowing users and authorities to verify the IMEI number of a device. Operators also check IMEI against the CEIR (Central Equipment Identity Register) to block stolen or blacklisted devices from being activated on new connections.

Request Structure

```
{
  "@type": "EvaluateFraudRisk",
  "requestedFraudEvaluationDate": "2025-03-16T11:30:00Z",
  "fraudTargetIdentifier": [
    {
      "@type": "PhoneContactMedium",
      "@baseType": "ContactMedium",
      "phoneNumber": "91-99887-76655"
    }
  ],
  "relatedParty": [
    {
      "@type": "RelatedPartyRefOrPartyRoleRef",
      "partyOrPartyRole": {
        "@type": "PartyRef",
        "id": "cust-ind-2025-088",
        "name": "Priya Sharma",
        "@referredType": "Individual"
      },
      "role": "Customer"
    }
  ],
  "fraudEvaluationCriteria": {
    "@type": "FraudEvaluationCriteria",
    "fraudRiskProfileCriteria": {
      "@type": "FraudRiskProfileCriteria",
      "deviceTenure": true,
      "simTenure": true,
      "serviceStatus": true,
      "lineTenure": false,
      "paymentMethod": false,
      "callForwarding": false,
      "simSwap": false
    },
    "fraudScoreCriteria": {
      "@type": "FraudScoreCriteria",
      "isExplanationRequired": true
    }
  }
```

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```
}
```

What Happens During Evaluation

1. Device Validation (fraudRiskProfileCriteria):

- Is the device IMEI recognized and valid?
- How long has this device been in use on the network? (A brand-new IMEI appearing with a used SIM may indicate a stolen device)
- What is the SIM tenure associated with this connection?
- Is the service currently active on this number?

2. CEIR / Sanchar Saathi Cross-Check (implicit in fraud scoring):

- The fraud scoring engine checks the device IMEI against the CEIR blacklist
- Stolen, lost, or cloned devices flagged in the central registry will result in a high fraud score
- Devices reported by users via the Sanchar Saathi portal are automatically reflected in CEIR

Response Example

```
{
  "@type": "EvaluateFraudRisk",
  "id": "imei-check-2025-001",
  "state": "done",
  "fraudEvaluationResult": {
    "@type": "FraudEvaluationResult",
    "fraudRiskProfileResult": {
      "@type": "FraudRiskProfileResult",
      "deviceTenure": {
        "units": "month",
        "amount": 0
      },
      "simTenure": {
        "units": "month",
        "amount": 24
      },
      "serviceStatus": "active"
    },
    "fraudScoreResult": {
      "@type": "FraudScoreResult",
      "score": "92",
      "explanation": "HIGH RISK - Device IMEI flagged in CEIR blacklist as stolen.
Device tenure is 0 months (newly inserted SIM into a different device). Recommend
blocking activation and alerting authorities."
    }
  }
```

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```
}
```

Interpretation

A very high fraud score (92) indicates the device is flagged in the CEIR blacklist. The device tenure of 0 months means the existing SIM (24 months old) was just inserted into this device, suggesting a legitimate subscriber may unknowingly be using a stolen handset, or a fraudster has obtained both a stolen device and a SIM. The operator should block the device activation and notify the subscriber and relevant authorities per Sanchar Saathi procedures.

TAFCOP — Connection Limit Check

Scenario: A telecom operator receives a new connection request from an individual. Before provisioning, the operator must verify through the TAFCOP (Telecom Analytics for Fraud Management and Consumer Protection) portal whether the applicant's identity documents have been used to register excessive mobile connections, which may indicate identity fraud or SIM farming.

Regulatory Context:

The Sanchar Saathi platform includes features to let users check if their names have been used to register additional mobile connections. Through the TAFCOP portal, operators and users can verify how many connections are registered against one identity, and flag or disconnect excess connections.

Request Structure

```
{
  "@type": "EvaluateFraudRisk",
  "requestedFraudEvaluationDate": "2025-03-16T14:00:00Z",
  "fraudTargetIdentifier": [
    {
      "@type": "PhoneContactMedium",
      "@baseType": "ContactMedium",
      "phoneNumber": "91-70001-23456"
    }
  ],
  "relatedParty": [
    {
      "@type": "RelatedPartyRefOrPartyRoleRef",
      "partyOrPartyRole": {
        "@type": "PartyRef",
        "id": "cust-ind-2025-112",
        "name": "Suresh Patel",
        "@referredType": "Individual"
      },
      "role": "Customer"
```

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```
      }
    ],
    "fraudEvaluationCriteria": {
      "@type": "FraudEvaluationCriteria",
      "fraudRiskProfileCriteria": {
        "@type": "FraudRiskProfileCriteria",
        "simTenure": false,
        "deviceTenure": false,
        "lineTenure": false,
        "paymentMethod": true,
        "callForwarding": false,
        "serviceStatus": false,
        "simSwap": false
      },
      "fraudScoreCriteria": {
        "@type": "FraudScoreCriteria",
        "isExplanationRequired": true
      },
      "fraudProfileMatchCriteria": {
        "@type": "FraudProfileMatchCriteria",
        "partyName": "Suresh Patel",
        "email": {
          "@type": "EmailContactMedium",
          "@baseType": "ContactMedium",
          "emailAddress": "suresh.patel99@email.com"
        },
        "address": {
          "@type": "GeographicAddressContactMedium",
          "@baseType": "ContactMedium",
          "city": "Ahmedabad",
          "country": "India",
          "postCode": "380015",
          "stateOrProvince": "Gujarat",
          "street1": "12, Navrangpura"
        }
      }
    }
  }
```

What Happens During Evaluation

1. Identity Cross-Referencing (fraudProfileMatchCriteria):

- Does the name, email, and address match existing records across operators?
- How many active connections are currently registered against this identity?
- Are there connections under slight name or address variations (fuzzy matching)?

2. Payment Method Check (fraudRiskProfileCriteria):

- What payment method is associated with existing connections? (SIM farms often use the same prepaid payment method across multiple connections)

3. TAFCOP Cross-Check (implicit in fraud scoring):

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- The fraud scoring engine queries TAFCOP data to determine the total number of connections registered to this identity
- Connections exceeding regulatory limits are flagged
- Previously disconnected or flagged connections under the same identity are factored into the score

Response Example


```
{
  "@type": "EvaluateFraudRisk",
  "id": "tafcop-check-2025-001",
  "state": "done",
  "fraudEvaluationResult": {
    "@type": "FraudEvaluationResult",
    "fraudRiskProfileResult": {
      "@type": "FraudRiskProfileResult",
      "paymentMethod": {
        "@type": "PaymentMethodRef",
        "id": "pm-prepaid-01",
        "name": "Prepaid",
        "@referredType": "Voucher"
      }
    },
    "fraudScoreResult": {
      "@type": "FraudScoreResult",
      "score": "85",
      "explanation": "HIGH RISK - TAFCOP records show 14 active connections registered
against this identity across multiple operators. Regulatory limit exceeded. 3
connections were previously flagged and disconnected for suspicious activity. All
connections use prepaid payment method. Recommend blocking new connection and
initiating identity verification."
    },
    "fraudProfileMatchResult": {
      "@type": "FraudProfileMatchResult",
      "globalMatchScore": "78",
      "profileMatch": [
        {
          "@type": "ProfileMatch",
          "matchElement": "name",
          "matchScore": "82"
        },
        {
          "@type": "ProfileMatch",
          "matchElement": "emailAddress",
          "matchScore": "70"
        },
        {
          "@type": "ProfileMatch",
          "matchElement": "address",
          "matchScore": "80"
        }
      ]
    }
  }
```


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```
}
```

Interpretation

A high fraud score (85) signals that the identity has excessive connections (14 active) registered across multiple operators, exceeding regulatory limits. The TAFCOP cross-check reveals 3 previously disconnected connections linked to suspicious activity. All connections use prepaid payment, a common pattern in SIM farming operations. The profile match score of 78 indicates moderate similarity with known flagged identities. The operator should block the new connection request and initiate a full identity verification process per TAFCOP guidelines.

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Support of polymorphism and extension patterns

Support of polymorphic collections and types and schema based extension is provided by means of a list of generic meta-attributes that we describe below. Polymorphism in collections occurs when entities inherit from base entities, for instance a BillingAccount and SettlementAccount inheriting properties from the abstract Account entity.

Generic support of polymorphism and pattern extensions is described in the TMF API Guidelines, Part 2 (TMF630).

The @type attribute provides a way to represent the actual class type of an entity. For example, within a list of Account instances some may be instances of BillingAccount where other could be instances of SettlementAccount. The @type gives this information. All resources and sub-resources of this API have a @type attributes that can be provided when this is useful.

The @referredType can be used within reference entities (like for instance an AccountRef object) to explicitly denote the actual entity type of the referred class. Notice that in reference entities the @type, when used, denotes the class type of the reference itself, such as BillingAccountRef or SettlementAccountRef, and not the class type of the referred object. However since reference classes are rarely sub-classed, @type is generally not useful in reference objects.

The @schemaLocation property can be used in resources to allow specifying user-defined properties of an Entity or to specify the expected characteristics of an entity.

The @baseType attribute gives a way to provide explicitly the base class of a given resource that has been extended through functional inheritance. Do not set @baseType to the same value as @type, or to technical classes such as Entity, EntityRef, Extensible, or Addressable. For this API's published examples, @baseType is populated only where that inheritance applies (for example `ContactMedium` for contact-medium subtypes, and `Event` for event subtypes). Use `Characteristic` as @baseType only when a concrete characteristic **subtype** extends `Characteristic` (not when @type is simply `Characteristic`).

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RESOURCE MODEL

Managed Entity and Task Resource Models

EvaluateFraudRisk resource

A task resource that coordinates and executes comprehensive fraud evaluation processes for telecommunications customers or transactions. It defines the criteria for fraud detection checks to be performed (such as risk profiling, network authentication, fraud scoring, and profile matching) and captures the results of these evaluations. The task identifies the target being assessed for fraud (typically a phone number or customer contact point) and associates relevant parties involved in the fraud assessment process.

Resource model

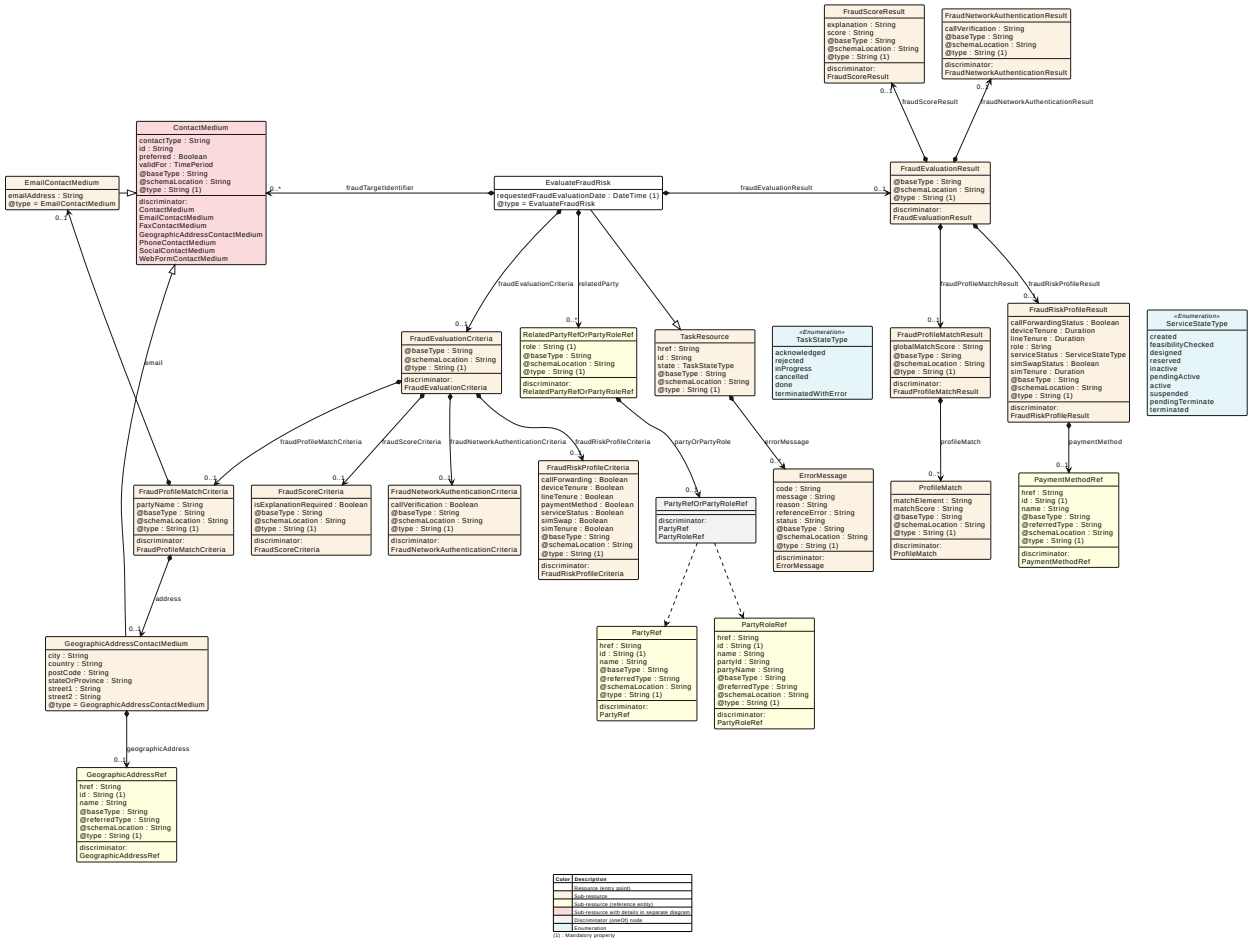


Figure 1 - EvaluateFraudRisk

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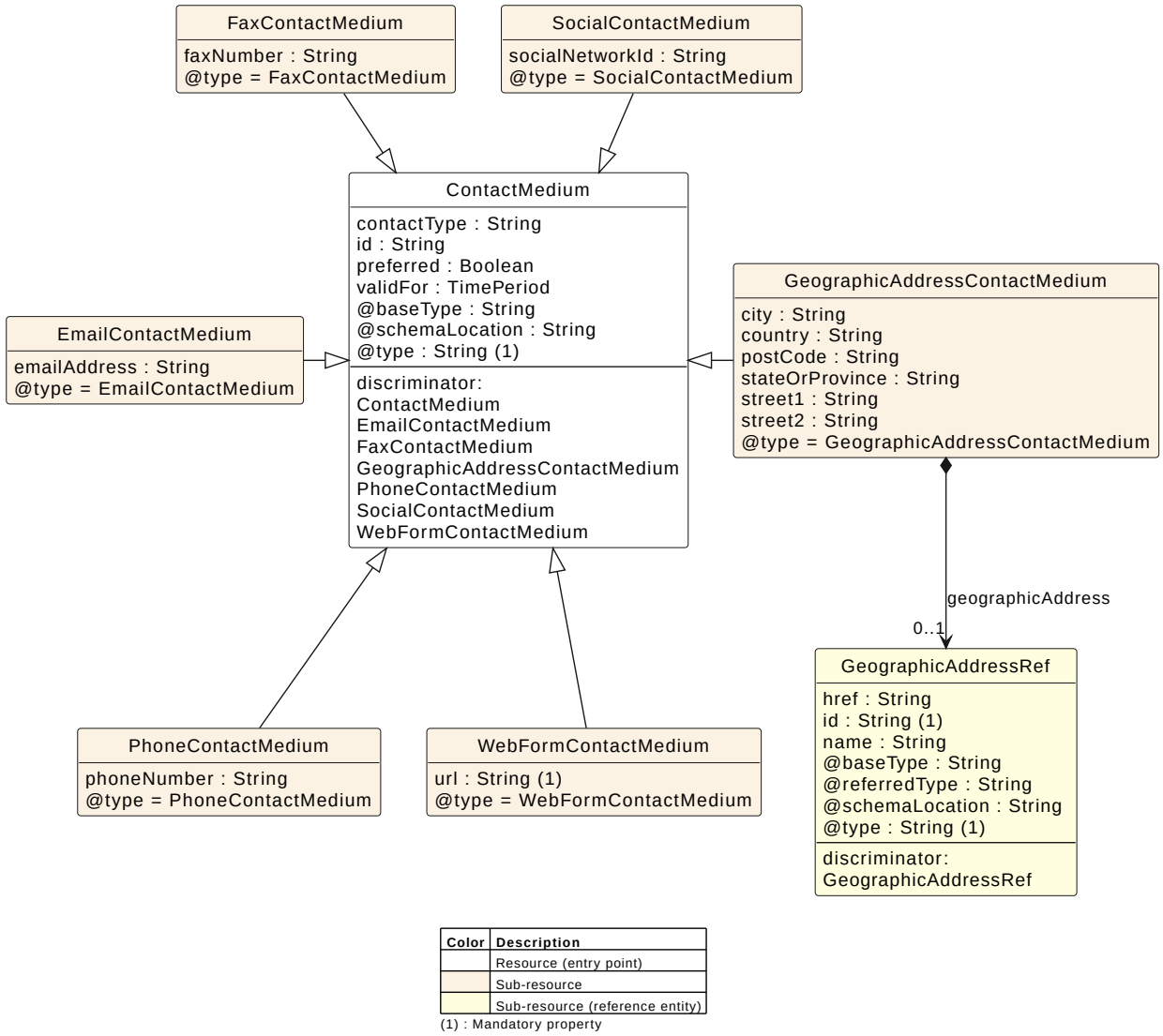


Figure 2 - ContactMedium

Field descriptions

EvaluateFraudRisk fields

errorMessage	An ErrorMessage [0..*]. A list of error message contains the error that caused the termination of the task.

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fraudEvaluationCriteria	A FraudEvaluationCriteria. Specifies the evaluation criteria and checks that should be performed during the fraud assessment process. This includes defining which fraud detection methods to apply, such as risk profile analysis (examining SIM/device tenure, payment history), network-level authentication verification, fraud scoring algorithms, and customer profile matching against known fraud patterns. Each criterion can be configured as required or optional based on the fraud evaluation requirements.
fraudEvaluationResult	A FraudEvaluationResult. Contains the comprehensive results from all fraud evaluation checks that were performed according to the specified criteria. This aggregates outcomes from multiple fraud detection methods including risk profile analysis results (tenure information, payment types, service status), network authentication verification status, calculated fraud risk scores with explanations, and profile matching results showing how closely the customer data matches known fraud patterns.
fraudTargetIdentifier	A ContactMedium [1..*]. Contact information identifying the target(s) being assessed for fraud (phone number, email, address, etc.). List allows multiple identifiers for the same target. Fraud evaluation is performed against each identifier; results may reveal correlations or discrepancies across contact mediums, helping detect identity fraud or account takeover. ContactMedium can be instantiated as * EmailContactMedium * FaxContactMedium * GeographicAddressContactMedium * PhoneContactMedium * SocialContactMedium * WebFormContactMedium
href	A String. Hyperlink reference.
id	A String. Unique identifier.

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relatedParty	A RelatedPartyRefOrPartyRoleRef [1..*]. Parties involved in the fraud evaluation. Mandatory for accountability and audit. Typically includes the party requesting the evaluation (role: 'requester' or 'initiator') and the party being assessed (role: 'merchant', 'customer', or 'subject').
requestedFraudEvaluationDate	A DateTime [1]. Date and time when the fraud evaluation was requested.
state	A TaskStateType. Possible values for the state of a task. ENUMERATED with values: * acknowledged * rejected * inProgress * cancelled * done * terminatedWithError
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

ErrorMessage sub-resource fields

Represents an Error.

code	A String. Error code.
message	A String. More details and corrective actions related to the error.
reason	A String. Explanation of the reason for the error.
referenceError	A String. URI of documentation describing the error.
status	A String. Error code extension like sys-ABC-2001.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

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FraudEvaluationCriteria sub-resource fields

Specifies the evaluation criteria and checks that should be performed during the fraud assessment process. This includes defining which fraud detection methods to apply, such as risk profile analysis (examining SIM/device tenure, payment history), network-level authentication verification, fraud scoring algorithms, and customer profile matching against known fraud patterns. Each criterion can be configured as required or optional based on the fraud evaluation requirements.

fraudNetworkAuthenticationCriteria	A FraudNetworkAuthenticationCriteria. Indicates if network authentication criteria is mandatory for fraud evaluation.
fraudProfileMatchCriteria	A FraudProfileMatchCriteria. Defines the customer profile information elements that should be used for matching analysis against known fraud patterns and watchlists. Specifies which customer identifiers and contact details (such as email address, name, and physical address) should be compared during the profile matching process to identify potential matches with fraudulent profiles or suspicious entities previously flagged in fraud databases.
fraudRiskProfileCriteria	A FraudRiskProfileCriteria. Defines whch customer and account risk profile factors should be evaluated during fraud detection analysis. Specifies whether to examine factors such as SIM card tenure, device tenure, line tenure, payment method type, call forwarding configuration, service activation status, and recent SIM swap history. Each factor can be selectively enabled or disabled to customize the risk assessment scope based on the fraud detection scenario.
fraudScoreCriteria	A FraudScoreCriteria. Defines the requirements and configuration for fraud risk scoring during the evaluation process. Specifies whether fraud scoring is mandatory for the evaluation and whether detailed explanations should be provided to describe the factors and reasoning behind the calculated fraud score. This allows customization of the fraud scoring depth and transparency based on business requirements or regulatory compliance needs.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.

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@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

FraudNetworkAuthenticationCriteria sub-resource fields

Indicates if network authentication criteria is mandatory for fraud evaluation.

callVerification	A Boolean. Indicates if call verification is considered in the network authentication criteria.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

FraudProfileMatchCriteria sub-resource fields

Defines the customer profile information elements that should be used for matching analysis against known fraud patterns and watchlists. Specifies which customer identifiers and contact details (such as email address, name, and physical address) should be compared during the profile matching process to identify potential matches with fraudulent profiles or suspicious entities previously flagged in fraud databases.

address	A GeographicAddressContactMedium. Describes a geographical address that could be used to contact a party (an individual or an organization).
email	An EmailContactMedium. Describes an email that could be used to contact a party (an individual or an organization).
partyName	A String. The actual party name (full name or individual name components) to be matched against fraud pattern databases. This value is used for fuzzy matching and similarity scoring to identify potential correlations with known fraudulent identities or suspicious name patterns.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

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GeographicAddressContactMedium sub-resource fields

Describes a geographical address that could be used to contact a party (an individual or an organization).

city	A String. The city.
contactType	A String. Type of the contact medium to qualify it like pro email / personal email. This is not used to define the contact medium used.
country	A String. The country.
geographicAddress	A GeographicAddressRef. Reference to a Geographic Address.
id	A String. Identifier for this contact medium.
postCode	A String. Postcode.
preferred	A Boolean. If true, indicates that is the preferred contact medium.
stateOrProvince	A String. State or province.
street1	A String. Describes the street.
street2	A String. Complementary street description.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

GeographicAddressRef sub-resource fields

Reference to a Geographic Address.

href	A String. The URI of the referred entity.
id	A String [1]. The identifier of the referred entity.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.

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@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

ContactMedium sub-resource fields

Indicates the contact medium that could be used to contact the party. This is an abstract base class, the actual value is in one of the strongly-typed subclasses : EmailContact Medium, FaxContactMedium, PhoneContactMedium, GeographicAddressContactMedium, SocialMediaContactMedium...

contactType	A String. Type of the contact medium to qualify it like pro email / personal email. This is not used to define the contact medium used.
id	A String. Identifier for this contact medium.
preferred	A Boolean. If true, indicates that is the preferred contact medium.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.
	ContactMedium can be instantiated as * EmailContactMedium * FaxContactMedium * GeographicAddressContactMedium * PhoneContactMedium * SocialContactMedium * WebFormContactMedium
value	This property is present in subclasses

EmailContactMedium sub-resource fields

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Describes an email that could be used to contact a party (an individual or an organization).

contactType	A String. Type of the contact medium to qualify it like pro email / personal email. This is not used to define the contact medium used.
emailAddress	A String. Full email address in standard format.
id	A String. Identifier for this contact medium.
preferred	A Boolean. If true, indicates that is the preferred contact medium.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

FaxContactMedium sub-resource fields

Describes a fax that could be used to contact a party (an individual or an organization).

contactType	A String. Type of the contact medium to qualify it like pro email / personal email. This is not used to define the contact medium used.
faxNumber	A String. The fax number of the contact.
id	A String. Identifier for this contact medium.
preferred	A Boolean. If true, indicates that is the preferred contact medium.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

PhoneContactMedium sub-resource fields

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Describes a phone number that could be used to contact a party (an individual or an organization).

contactType	A String. Type of the contact medium to qualify it like pro email / personal email. This is not used to define the contact medium used.
id	A String. Identifier for this contact medium.
phoneNumber	A String. The phone number of the contact.
preferred	A Boolean. If true, indicates that is the preferred contact medium.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

SocialContactMedium sub-resource fields

Describes a social media identifier that could be used to contact a party (an individual or an organization).

contactType	A String. Type of the contact medium to qualify it like pro email / personal email. This is not used to define the contact medium used.
id	A String. Identifier for this contact medium.
preferred	A Boolean. If true, indicates that is the preferred contact medium.
socialNetworkId	A String. Identifier as a member of a social network.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.

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@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

WebFormContactMedium sub-resource fields

Describes an URL of a web form that could be used to contact a party (an individual or an organization).

contactType	A String. Type of the contact medium to qualify it like pro email / personal email. This is not used to define the contact medium used.
id	A String. Identifier for this contact medium.
preferred	A Boolean. If true, indicates that is the preferred contact medium.
url	A String [1]. Url of the web form.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

FraudRiskProfileCriteria sub-resource fields

Defines whch customer and account risk profile factors should be evaluated during fraud detection analysis. Specifies whether to examine factors such as SIM card tenure, device tenure, line tenure, payment method type, call forwarding configuration, service activation status, and recent SIM swap history. Each factor can be selectively enabled or disabled to customize the risk assessment scope based on the fraud detection scenario.

callForwarding	A Boolean. Indicates if call forwarding is considered in the criteria.
deviceTenure	A Boolean. Indicates if device tenure is considered in the criteria.
lineTenure	A Boolean. Indicates if line tenure is considered in the criteria.

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paymentMethod	A Boolean. Indicates if payment method information is considered in the fraud risk evaluation criteria.
serviceStatus	A Boolean. Indicates if service status is considered in the criteria.
simSwap	A Boolean. Check if number has gone through recent simswap.
simTenure	A Boolean. Indicates if SIM tenure is considered in the criteria.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

FraudScoreCriteria sub-resource fields

Defines the requirements and configuration for fraud risk scoring during the evaluation process. Specifies whether fraud scoring is mandatory for the evaluation and whether detailed explanations should be provided to describe the factors and reasoning behind the calculated fraud score. This allows customization of the fraud scoring depth and transparency based on business requirements or regulatory compliance needs.

isExplanationRequired	A Boolean. Indicates if an explanation is required for the fraud score.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

FraudEvaluationResult sub-resource fields

Contains the comprehensive results from all fraud evaluation checks that were performed according to the specified criteria. This aggregates outcomes from multiple fraud detection methods including risk profile analysis results (tenure information, payment types, service status), network authentication verification status, calculated fraud risk scores with explanations, and profile matching results showing how closely the customer data matches known fraud patterns.

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fraudNetworkAuthenticationResult	A FraudNetworkAuthenticationResult. Contains the results of network authentication verification checks performed during fraud evaluation. Provides the authentication status indicating whether the network connection and calling party have been successfully verified through telecommunications network authentication mechanisms, helping to identify potential spoofing or unauthorized access attempts.
fraudProfileMatchResult	A FraudProfileMatchResult. Contains the results of customer profile matching analysis performed against fraud pattern databases and watchlists. Provides an overall global match score indicating the strength of correlation with known fraud profiles, along with detailed matching results for individual profile elements (such as name, email, address) showing which specific data points matched and their individual confidence scores. This helps identify customers or transactions that closely resemble known fraudulent patterns.
fraudRiskProfileResult	A FraudRiskProfileResult. Contains the actual values and status information for risk profile factors retrieved from the telecommunications system during fraud evaluation. Provides concrete data such as tenure durations (for SIM, device, and line expressed as Duration with flexible time units), payment method registered with the telco, call forwarding settings, current service status, account holder role, and recent SIM swap activity. This information is used to assess the fraud risk level based on customer account characteristics and usage patterns.
fraudScoreResult	A FraudScoreResult. Contains the calculated fraud risk score and supporting information from the fraud scoring algorithm. Provides a numerical score indicating the level of fraud risk detected (typically on a scale like 0-100, where higher values indicate greater fraud risk) along with an optional textual explanation describing the key factors that influenced the score, such as credit history, transaction patterns, or behavioral anomalies.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.

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@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

FraudNetworkAuthenticationResult sub-resource fields

Contains the results of network authentication verification checks performed during fraud evaluation. Provides the authentication status indicating whether the network connection and calling party have been successfully verified through telecommunications network authentication mechanisms, helping to identify potential spoofing or unauthorized access attempts.

callVerification	A String. Result of telecom call verification where the customer is required to enter a pre-shared authentication code received via OBD (outbound dialing) call or SMS to confirm their identity.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

FraudProfileMatchResult sub-resource fields

Contains the results of customer profile matching analysis performed against fraud pattern databases and watchlists. Provides an overall global match score indicating the strength of correlation with known fraud profiles, along with detailed matching results for individual profile elements (such as name, email, address) showing which specific data points matched and their individual confidence scores. This helps identify customers or transactions that closely resemble known fraudulent patterns.

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globalMatchScore	A String. Overall aggregated match score indicating the strength of correlation between the customer profile and known fraud patterns, typically expressed as a string to accommodate various scoring systems (percentage like '90', decimal like '0.90', categorical like 'HIGH', or qualified ranges like '85-95'). The calculation methodology may vary by implementation but commonly uses weighted averaging, maximum value, or composite algorithms based on individual profileMatch element scores. For example, if profileMatch contains email:'95', name:'85', address:'80', the global score might be '90' (weighted average) or '95' (highest match) depending on the fraud detection system's algorithm.
profileMatch	A ProfileMatch [0..*]. Array of profile matches.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

ProfileMatch sub-resource fields

Represents the matching analysis result for a single customer profile element during fraud detection. Identifies which specific profile attribute (such as name, email, phone number, or address) was evaluated and provides a numerical score indicating how closely it matches against reference fraud patterns or watchlist entries. Higher scores indicate stronger matches and potentially higher fraud risk for that particular profile element.

matchElement	A String. The specific customer profile attribute being evaluated for matching against fraud patterns. Common examples include 'name', 'email', 'address', 'phoneNumber', or other identifying customer information that can be compared against fraud databases or watchlists.

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matchScore	A String. Match confidence score for this profile element, expressed as a string to support multiple scoring formats used by different fraud detection systems. Common formats include: percentage strings ('90', '85.5'), decimal confidence scores ('0.90'), categorical ratings ('HIGH', 'MEDIUM', 'LOW'), or qualified scores ('90-EXACT', '75-FUZZY'). String type provides flexibility for fraud detection vendors to use their native scoring systems while maintaining API compatibility. Numerical percentage interpretation: 0-100 where higher values indicate stronger matches and greater fraud likelihood, with 100 representing a perfect match.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

FraudRiskProfileResult sub-resource fields

Contains the actual values and status information for risk profile factors retrieved from the telecommunications system during fraud evaluation. Provides concrete data such as tenure durations (for SIM, device, and line expressed as Duration with flexible time units), payment method registered with the telco, call forwarding settings, current service status, account holder role, and recent SIM swap activity. This information is used to assess the fraud risk level based on customer account characteristics and usage patterns.

callForwardingStatus	A Boolean. Call forwarding status.
deviceTenure	A Duration. A time interval in a given unit of time.
lineTenure	A Duration. A time interval in a given unit of time.
paymentMethod	A PaymentMethodRef. PaymentMethod reference. A payment method defines a specific mean of payment (e.g direct debit).
role	A String. The role or permission level of the account holder within the service subscription (e.g., 'Admin', 'User', 'Owner', 'Authorized User'). Account roles with elevated privileges may present different fraud risk profiles than standard user accounts.

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serviceStatus	A ServiceStateType. Valid values for the lifecycle state of the service. ENUMERATED with values: * created * feasibilityChecked * designed * reserved * inactive * pendingActive * active * suspended * pendingTerminate * terminated
simSwapStatus	A Boolean. Recent simswap status.
simTenure	A Duration. A time interval in a given unit of time.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

PaymentMethodRef sub-resource fields

PaymentMethod reference. A payment method defines a specific mean of payment (e.g direct debit).

href	A String. The URI of the referred entity.
id	A String [1]. The identifier of the referred entity.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

FraudScoreResult sub-resource fields

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Contains the calculated fraud risk score and supporting information from the fraud scoring algorithm. Provides a numerical score indicating the level of fraud risk detected (typically on a scale like 0-100, where higher values indicate greater fraud risk) along with an optional textual explanation describing the key factors that influenced the score, such as credit history, transaction patterns, or behavioral anomalies.

explanation	A String. Explanation for the fraud score.
score	A String. Score related to fraud evaluation.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

RelatedPartyRefOrPartyRoleRef sub-resource fields

RelatedParty reference. A related party defines party or party role or its reference, linked to a specific entity.

partyOrPartyRole	A PartyRefOrPartyRoleRef.
role	A String [1]. Role played by the related party or party role in the context of the specific entity it is linked to. Such as 'initiator', 'customer', 'salesAgent', 'user'.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. When sub-classing, this defines the sub-class Extensible name.

PartyRef sub-resource fields

A Party reference.

href	A String. The URI of the referred entity.
id	A String [1]. The identifier of the referred entity.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.

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@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. ENUMERATED with values: * PartyRef

PartyRoleRef sub-resource fields

Party role reference. A party role represents the part played by a party in a given context.

href	A String. The URI of the referred entity.
id	A String [1]. The identifier of the referred entity.
name	A String. Name of the referred entity.
partyId	A String. The identifier of the engaged party that is linked to the PartyRole object.
partyName	A String. The name of the engaged party that is linked to the PartyRole object.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String [1]. ENUMERATED with values: * PartyRoleRef

Json representation sample(s)

We provide below a JSON representation as example of the 'EvaluateFraudRisk' resource object.


```
{
  "@type": "EvaluateFraudRisk",
  "@schemaLocation": "/schema-registry/EvaluateFraudRisk.yaml",
  "href": "/tmf-api/fraudManagement/v5/evaluateFraudRisk/f3a34f89-0046-tsk-01",
  "id": "f3a34f89-0046-tsk-01",
  "requestedFraudEvaluationDate": "2024-11-20T09:34:58Z",
  "state": "done",
  "fraudTargetIdentifier": [
    {
      "@type": "PhoneContactMedium",
```


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```
      "@baseType": "ContactMedium",
      "@schemaLocation": "/schema-registry/PhoneContactMedium.yaml",
      "phoneNumber": "01 09 75 83 51"
    }
  ],
  "relatedParty": [
    {
      "@type": "RelatedPartyRefOrPartyRoleRef",
      "@schemaLocation": "/schema-registry/RelatedPartyRef.yaml",
      "partyOrPartyRole": {
        "@type": "PartyRef",
        "href": "/tmf-api/partyManagement/v5/Organization/f3a34f89-0046-pty-01",
        "id": "f3a34f89-0046-pty-01",
        "name": "Bank XYZ",
        "@referredType": "Organization"
      },
      "role": "Merchant"
    }
  ],
  "fraudEvaluationCriteria": {
    "@type": "FraudEvaluationCriteria",
    "@schemaLocation": "/schema-registry/FraudEvaluationCriteria.yaml",
    "fraudRiskProfileCriteria": {
      "@type": "FraudRiskProfileCriteria",
      "@schemaLocation": "/schema-registry/FraudRiskProfileCriteria.yaml",
      "simTenure": false,
      "deviceTenure": true,
      "lineTenure": false,
      "paymentMethod": false,
      "callForwarding": true,
      "serviceStatus": true,
      "simSwap": false
    },
    "fraudNetworkAuthenticationCriteria": {
      "@type": "FraudNetworkAuthenticationCriteria",
      "@schemaLocation": "/schema-registry/FraudNetworkAuthenticationCriteria.yaml",
      "callVerification": false
    },
    "fraudScoreCriteria": {
      "@type": "FraudScoreCriteria",
      "@schemaLocation": "/schema-registry/FraudScoreCriteria.yaml",
      "isExplanationRequired": true
    },
    "fraudProfileMatchCriteria": {
      "@type": "FraudProfileMatchCriteria",
      "@schemaLocation": "/schema-registry/FraudProfileMatchCriteria.yaml",
      "email": {
        "@type": "EmailContactMedium",
        "@baseType": "ContactMedium",
        "@schemaLocation": "/schema-registry/EmailContactMedium.yaml",
        "emailAddress": "neo@matrix.com"
      },
      "partyName": "Joe Doe",
      "address": {
        "@type": "GeographicAddressContactMedium",
        "@baseType": "ContactMedium",
        "@schemaLocation": "/schema-registry/GeographicAddressContactMedium.yaml",
        "id": "address-001",
        "preferred": true,
        "contactType": "private",
```


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```
        "validFor": {
          "startDateTime": "2025-01-29T10:40:52Z",
          "endDateTime": "2025-07-29T10:40:52Z"
        },
        "city": "Gurgaon",
        "country": "India",
        "postCode": "121001",
        "stateOrProvince": "Harayana",
        "street1": "Sector 39",
        "street2": "Near Unitech cyber park",
        "geographicAddress": {
          "@type": "GeographicAddressRef",
          "@schemaLocation": "/schema-registry/GeographicAddressRef.yaml",
          "href": "/tmf-api/geographicAddressManagement/v5/geographicAddress/35dcfeec-
9051-4b05-830e7a0f67dc541d",
          "id": "35dcfeec-9051-4b05-830e7a0f67dc541d",
          "name": "address1",
          "@referredType": "GeographicAddress"
        }
      }
    }
  },
  "fraudEvaluationResult": {
    "@type": "FraudEvaluationResult",
    "@schemaLocation": "/schema-registry/FraudEvaluationResult.yaml",
    "fraudRiskProfileResult": {
      "@type": "FraudRiskProfileResult",
      "@schemaLocation": "/schema-registry/FraudRiskProfileResult.yaml",
      "simTenure": {
        "units": "month",
        "amount": 120
      },
      "deviceTenure": {
        "units": "month",
        "amount": 48
      },
      "lineTenure": {
        "units": "month",
        "amount": 120
      },
      "paymentMethod": {
        "@type": "PaymentMethodRef",
        "id": "pm-cc-01",
        "name": "Credit Card",
        "@referredType": "BankCard"
      },
      "callForwardingStatus": true,
      "serviceStatus": "active",
      "role": "Admin",
      "simSwapStatus": false
    },
    "fraudNetworkAuthenticationResult": {
      "@type": "FraudNetworkAuthenticationResult",
      "@schemaLocation": "/schema-registry/FraudNetworkAuthenticationResult.yaml",
      "callVerification": "VERIFIED"
    },
    "fraudScoreResult": {
      "@type": "FraudScoreResult",
      "@schemaLocation": "/schema-registry/FraudScoreResult.yaml",
      "score": "70",
```


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```
      "explanation": "Credit Score low"
    },
    "fraudProfileMatchResult": {
      "@type": "FraudProfileMatchResult",
      "globalMatchScore": "90",
      "profileMatch": [
        {
          "@type": "ProfileMatch",
          "matchElement": "name",
          "matchScore": "90"
        }
      ]
    }
  }
}
```


Notification Resource Models

2 notifications are defined for this API.

Notifications related to EvaluateFraudRisk:

- Create Event
- State Change Event

The notification structure for all notifications in this API follow the pattern depicted by the figure below. A notification event resource (depicted by "SpecificEvent" placeholder) is a sub class of a generic Event structure containing at least an id of the event occurrence (eventId), an event timestamp (eventTime), and the name of the resource (eventType). This notification structure owns an event payload structure ("SpecificEventPayload" placeholder) linked to the resource concerned by the notification using the resource name as access field ("resourceName" placeholder).

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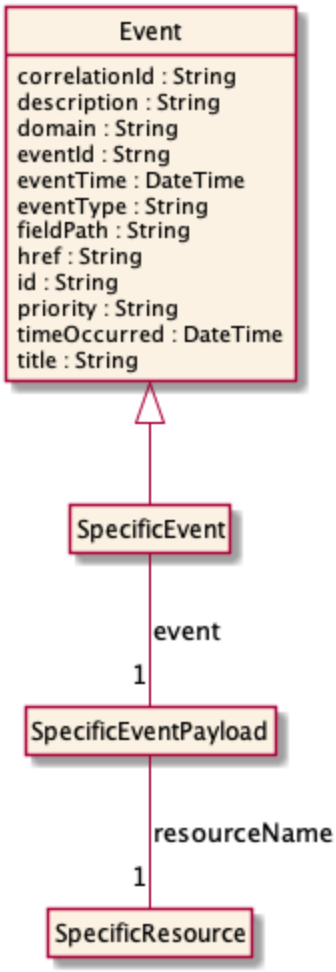


Figure 3 Notification Pattern

EvaluateFraudRisk

Create Event

Notification sent when a new fraud risk evaluation task has been created

```
Content-Type: application/json
{
  "@type": "EvaluateFraudRiskCreateEvent",
  "@baseType": "Event",
  "@schemaLocation": "/schema-registry/EvaluateFraudRiskCreateEvent.yaml",
  "href": "/tmf-api/fraudManagement/v5/hub/f3a34f89-0046-ar-01",
  "id": "f3a34f89-0046-ar-01",
  "correlationId": "f3a34f89-0046-cr-01",
  "domain": "Commercial",
  "title": "EvaluateFraudRiskCreateEvent",
  "description": "EvaluateFraudRiskCreateEvent illustration",
  "priority": "5",
  "timeOccurred": "2024-11-18T09:35:14Z",
  "source": {
    "@type": "ResourceRef",
```

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```
    "@schemaLocation": "/schema-registry/EntityRef.yaml",
    "href": "/tmf-api/resourceInventoryManagement/v5/logicalResource/f3a34f89-0046-ts-
01",
    "id": "f3a34f89-0046-ts-01",
    "name": "CRM App",
    "@referredType": "LogicalResource"
  },
  "reportingSystem": {
    "@type": "ResourceRef",
    "@schemaLocation": "/schema-registry/EntityRef.yaml",
    "href": "/tmf-api/resourceInventoryManagement/v5/logicalResource/f3a34f89-0046-ds-
01",
    "id": "f3a34f89-0046-ds-01",
    "name": "APP-877",
    "@referredType": "LogicalResource"
  },
  "relatedParty": [
    {
      "@type": "RelatedPartyRefOrPartyRoleRef",
      "@schemaLocation": "/schema-registry/RelatedPartyRefOrPartyRoleRef.yaml",
      "role": "initiator",
      "partyOrPartyRole": {
        "@type": "PartyRoleRef",
        "@schemaLocation": "/schema-registry/PartyRoleRef.yaml",
        "href": "/tmf-api/customerManagement/v5/customer/f3a34f89-0046-nn-01",
        "id": "f3a34f89-0046-nn-01",
        "name": "John Doe",
        "@referredType": "Customer",
        "partyId": "f3a34f89-0046-pt-01",
        "partyName": "John Doe"
      }
    },
    {
      "@type": "RelatedPartyRefOrPartyRoleRef",
      "@schemaLocation": "/schema-registry/RelatedPartyRefOrPartyRoleRef.yaml",
      "role": "initiator",
      "partyOrPartyRole": {
        "@type": "PartyRef",
        "@schemaLocation": "/schema-registry/PartyRef.yaml",
        "href": "/tmf-api/partyManagement/v5/organization/f3a34f89-a046-ar-01",
        "id": "f3a34f89-a046-ar-01",
        "name": "6d Tech",
        "@referredType": "Organization"
      }
    }
  ],
  "eventId": "f3a34f89-0046-evtid-01",
  "eventTime": "2024-11-18T09:35:14Z",
  "eventType": "EvaluateFraudRiskCreateEvent",
  "event": {
    "evaluateFraudRisk": {
      "@type": "EvaluateFraudRisk",
      "@schemaLocation": "/schema-registry/EvaluateFraudRisk.yaml",
      "href": "/tmf-api/fraudManagement/v5/evaluateFraudRisk/f3a34f89-0046-tsk-01",
      "id": "f3a34f89-0046-tsk-01",
      "requestedFraudEvaluationDate": "2024-11-20T09:34:58Z",
      "state": "inProgress",
      "fraudTargetIdentifier": [
        {
          "@type": "PhoneContactMedium",
```


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```
          "@baseType": "ContactMedium",
          "@schemaLocation": "/schema-registry/PhoneContactMedium.yaml",
          "phoneNumber": "01 09 75 83 51"
        }
      ],
      "relatedParty": [
        {
          "@type": "RelatedPartyRefOrPartyRoleRef",
          "@schemaLocation": "/schema-registry/RelatedPartyRef.yaml",
          "partyOrPartyRole": {
            "@type": "PartyRef",
            "href": "/tmf-api/partyManagement/v5/Organization/f3a34f89-0046-pty-01",
            "id": "f3a34f89-0046-pty-01",
            "name": "Bank XYZ",
            "@referredType": "Organization"
          },
          "role": "Merchant"
        }
      ],
      "fraudEvaluationCriteria": {
        "@type": "FraudEvaluationCriteria",
        "@schemaLocation": "/schema-registry/FraudEvaluationCriteria.yaml",
        "fraudRiskProfileCriteria": {
          "@type": "FraudRiskProfileCriteria",
          "@schemaLocation": "/schema-registry/FraudRiskProfileCriteria.yaml",
          "simTenure": false,
          "deviceTenure": true,
          "lineTenure": false,
          "paymentMethod": false,
          "callForwarding": true,
          "serviceStatus": true,
          "simSwap": false
        },
        "fraudNetworkAuthenticationCriteria": {
          "@type": "FraudNetworkAuthenticationCriteria",
          "@schemaLocation": "/schema-
registry/FraudNetworkAuthenticationCriteria.yaml",
          "callVerification": false
        },
        "fraudScoreCriteria": {
          "@type": "FraudScoreCriteria",
          "@schemaLocation": "/schema-registry/FraudScoreCriteria.yaml",
          "isExplanationRequired": true
        },
        "fraudProfileMatchCriteria": {
          "@type": "FraudProfileMatchCriteria",
          "@schemaLocation": "/schema-registry/FraudProfileMatchCriteria.yaml",
          "email": {
            "@type": "EmailContactMedium",
            "@baseType": "ContactMedium",
            "@schemaLocation": "/schema-registry/EmailContactMedium.yaml",
            "emailAddress": "neo@matrix.com"
          },
          "partyName": "Joe Doe",
          "address": {
            "@type": "GeographicAddressContactMedium",
            "@baseType": "ContactMedium",
            "@schemaLocation": "/schema-registry/GeographicAddressContactMedium.yaml",
            "id": "address-001",
            "preferred": true,
```


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```
            "contactType": "private",
            "validFor": {
              "startDateTime": "2025-01-29T10:40:52Z",
              "endDateTime": "2025-07-29T10:40:52Z"
            },
            "city": "Gurgaon",
            "country": "India",
            "postCode": "121001",
            "stateOrProvince": "Harayana",
            "street1": "Sector 39",
            "street2": "Near Unitech cyber park",
            "geographicAddress": {
              "@type": "GeographicAddressRef",
              "@schemaLocation": "/schema-registry/GeographicAddressRef.yaml",
              "href": "/tmf-
api/geographicAddressManagement/v5/geographicAddress/35dcfeec-9051-4b05-
830e7a0f67dc541d",
              "id": "35dcfeec-9051-4b05-830e7a0f67dc541d",
              "name": "address1",
              "@referredType": "GeographicAddress"
            }
          }
        }
      }
    }
  }
}
```


State Change Event

Notification sent when a fraud risk evaluation task changes state (e.g., from in-progress to completed or cancelled)


```
Content-Type: application/json
{
  "@type": "EvaluateFraudRiskStateChangeEvent",
  "@baseType": "Event",
  "@schemaLocation": "/schema-registry/EvaluateFraudRiskStateChangeEvent.yaml",
  "href": "/tmf-api/fraudManagement/v5/hub/f3a34f89-0046-ar-01",
  "id": "f3a34f89-0046-01",
  "correlationId": "f3a34f89-0046-cr-01",
  "domain": "Commercial",
  "title": "EvaluateFraudRiskStateChangeEvent",
  "description": "EvaluateFraudRiskStateChangeEvent illustration",
  "priority": "5",
  "timeOccurred": "2024-11-18T09:35:14Z",
  "source": {
    "@type": "EntityRef",
    "@schemaLocation": "/schema-registry/EntityRef.yaml",
    "href": "/tmf-api/resourceInventoryManagement/v5/logicalResource/f3a34f89-0046-src-
01",
    "id": "f3a34f89-0046-src-01",
    "name": "CRM App",
    "@referredType": "LogicalResource"
  },
  "reportingSystem": {
```


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```
    "@type": "EntityRef",
    "@schemaLocation": "/schema-registry/EntityRef.yaml",
    "href": "/tmf-api/resourceInventoryManagement/v5/logicalResource/f3a34f89-0046-rs-
01",
    "id": "f3a34f89-0046-rs-01",
    "name": "APP-877",
    "@referredType": "LogicalResource"
  },
  "relatedParty": [
    {
      "@type": "RelatedPartyRefOrPartyRoleRef",
      "@schemaLocation": "/schema-registry/RelatedPartyRefOrPartyRoleRef.yaml",
      "role": "initiator",
      "partyOrPartyRole": {
        "@type": "PartyRoleRef",
        "@schemaLocation": "/schema-registry/PartyRoleRef.yaml",
        "href": "/tmf-api/customerManagement/v5/customer/f3a34f89-0046-pt-51",
        "id": "f3a34f89-0046-pt-51",
        "name": "John Doe",
        "@referredType": "Customer",
        "partyId": "f3a34f89-0046-pt-01",
        "partyName": "John Doe"
      }
    },
    {
      "@type": "RelatedPartyRefOrPartyRoleRef",
      "@schemaLocation": "/schema-registry/RelatedPartyRefOrPartyRoleRef.yaml",
      "role": "initiator",
      "partyOrPartyRole": {
        "@type": "PartyRef",
        "@schemaLocation": "/schema-registry/PartyRef.yaml",
        "href": "/tmf-api/partyManagement/v5/organization/f3a34f89-0046-pt-01",
        "id": "f3a34f89-0046-pt-01",
        "name": "6d Tech",
        "@referredType": "Organization"
      }
    }
  ],
  "eventId": "f3a34f89-0046-evtid-01",
  "eventTime": "2024-11-18T09:35:14Z",
  "eventType": "EvaluateFraudRiskStateChangeEvent",
  "event": {
    "evaluateFraudRisk": {
      "id": "f3a34f89-0046-tsk-01",
      "href": "/tmf-api/fraudManagement/v5/evaluateFraudRisk/f3a34f89-0046-tsk-01",
      "@type": "EvaluateFraudRisk",
      "state": "done"
    }
  }
}
```


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API OPERATIONS

Remember the following Uniform Contract:

Operation on Entities	Uniform API Operation	Description
Query Entities	GET Resource	GET must be used to retrieve a representation of a resource.
Create Entity	POST Resource	POST must be used to create a new resource
Partial Update of an Entity	PATCH Resource	PATCH must be used to partially update a resource
Remove an Entity	DELETE Resource	DELETE must be used to remove a resource
Execute an Action on an Entity	POST on TASK Resource	POST must be used to execute Task Resources
Other Request Methods	POST on TASK Resource	GET and POST must not be used to tunnel other request methods.

Filtering and attribute selection rules are described in the TMF REST Design Guidelines.

Notifications are also described in a subsequent section.

Operations on EvaluateFraudRisk

Retrieves a evaluateFraudRisk by ID

`GET /evaluateFraudRisk/{id}?fields=…`

Description

This operation retrieves an EvaluateFraudRisk entity. Attribute selection is enabled for all first level attributes.

Usage samples

Retrieve details of a specific fraud risk evaluation task by its identifier

Request

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```
GET /evaluateFraudRisk/f3a34f89-0046-tsk-01
Content-Type: application/json
```

Response

```
200

{
  "@type": "EvaluateFraudRisk",
  "@schemaLocation": "/schema-registry/EvaluateFraudRisk.yaml",
  "href": "/tmf-api/fraudManagement/v5/evaluateFraudRisk/f3a34f89-0046-tsk-01",
  "id": "f3a34f89-0046-tsk-01",
  "requestedFraudEvaluationDate": "2024-11-20T09:34:58Z",
  "state": "done",
  "fraudTargetIdentifier": [
    {
      "@type": "PhoneContactMedium",
      "@baseType": "ContactMedium",
      "@schemaLocation": "/schema-registry/PhoneContactMedium.yaml",
      "phoneNumber": "01 09 75 83 51"
    }
  ],
  "relatedParty": [
    {
      "@type": "RelatedPartyRefOrPartyRoleRef",
      "@schemaLocation": "/schema-registry/RelatedPartyRef.yaml",
      "partyOrPartyRole": {
        "@type": "PartyRef",
        "href": "/tmf-api/partyManagement/v5/Organization/f3a34f89-0046-pty-01",
        "id": "f3a34f89-0046-pty-01",
        "name": "Bank XYZ",
        "@referredType": "Organization"
      },
      "role": "Merchant"
    }
  ],
  "fraudEvaluationCriteria": {
    "@type": "FraudEvaluationCriteria",
    "@schemaLocation": "/schema-registry/FraudEvaluationCriteria.yaml",
    "fraudRiskProfileCriteria": {
      "@type": "FraudRiskProfileCriteria",
      "@schemaLocation": "/schema-registry/FraudRiskProfileCriteria.yaml",
      "simTenure": false,
      "deviceTenure": true,
      "lineTenure": false,
      "paymentMethod": false,
      "callForwarding": true,
      "serviceStatus": true,
      "simSwap": false
    },
    "fraudNetworkAuthenticationCriteria": {
      "@type": "FraudNetworkAuthenticationCriteria",
      "@schemaLocation": "/schema-registry/FraudNetworkAuthenticationCriteria.yaml",
      "callVerification": false
    },
    "fraudScoreCriteria": {
```

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```
      "@type": "FraudScoreCriteria",
      "@schemaLocation": "/schema-registry/FraudScoreCriteria.yaml",
      "isExplanationRequired": true
    },
    "fraudProfileMatchCriteria": {
      "@type": "FraudProfileMatchCriteria",
      "@schemaLocation": "/schema-registry/FraudProfileMatchCriteria.yaml",
      "email": {
        "@type": "EmailContactMedium",
        "@baseType": "ContactMedium",
        "@schemaLocation": "/schema-registry/EmailContactMedium.yaml",
        "emailAddress": "neo@matrix.com"
      },
      "partyName": "Joe Doe",
      "address": {
        "@type": "GeographicAddressContactMedium",
        "@baseType": "ContactMedium",
        "@schemaLocation": "/schema-registry/GeographicAddressContactMedium.yaml",
        "id": "address-001",
        "preferred": true,
        "contactType": "private",
        "validFor": {
          "startDateTime": "2025-01-29T10:40:52Z",
          "endDateTime": "2025-07-29T10:40:52Z"
        },
        "city": "Gurgaon",
        "country": "India",
        "postCode": "121001",
        "stateOrProvince": "Harayana",
        "street1": "Sector 39",
        "street2": "Near Unitech cyber park",
        "geographicAddress": {
          "@type": "GeographicAddressRef",
          "@schemaLocation": "/schema-registry/GeographicAddressRef.yaml",
          "href": "/tmf-api/geographicAddressManagement/v5/geographicAddress/35dcfeec-
9051-4b05-830e7a0f67dc541d",
          "id": "35dcfeec-9051-4b05-830e7a0f67dc541d",
          "name": "address1",
          "@referredType": "GeographicAddress"
        }
      }
    }
  },
  "fraudEvaluationResult": {
    "@type": "FraudEvaluationResult",
    "@schemaLocation": "/schema-registry/FraudEvaluationResult.yaml",
    "fraudRiskProfileResult": {
      "@type": "FraudRiskProfileResult",
      "@schemaLocation": "/schema-registry/FraudRiskProfileResult.yaml",
      "simTenure": {
        "units": "month",
        "amount": 120
      },
      "deviceTenure": {
        "units": "month",
        "amount": 48
      },
      "lineTenure": {
        "units": "month",
        "amount": 120
```


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```
      },
      "paymentMethod": {
        "@type": "PaymentMethodRef",
        "id": "pm-cc-01",
        "name": "Credit Card",
        "@referredType": "BankCard"
      },
      "callForwardingStatus": true,
      "serviceStatus": "active",
      "role": "Admin",
      "simSwapStatus": false
    },
    "fraudNetworkAuthenticationResult": {
      "@type": "FraudNetworkAuthenticationResult",
      "@schemaLocation": "/schema-registry/FraudNetworkAuthenticationResult.yaml",
      "callVerification": "VERIFIED"
    },
    "fraudScoreResult": {
      "@type": "FraudScoreResult",
      "@schemaLocation": "/schema-registry/FraudScoreResult.yaml",
      "score": "70",
      "explanation": "Credit Score low"
    },
    "fraudProfileMatchResult": {
      "@type": "FraudProfileMatchResult",
      "globalMatchScore": "90",
      "profileMatch": [
        {
          "@type": "ProfileMatch",
          "matchElement": "name",
          "matchScore": "90"
        }
      ]
    }
  }
}
```


List or find evaluateFraudRisk objects

`GET /evaluateFraudRisk?fields=…&{filtering}`

Description

This operation list EvaluateFraudRisk entities. Attribute selection is enabled for all first level attributes. Filtering may be available depending on the compliance level supported by an implementation.

Usage samples

Retrieve a list of fraud risk evaluation tasks filtered by state (done)

Request

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```
GET /evaluateFraudRisk?filter=state=done
Content-Type: application/json
```

Response

```
200

[
  {
    "@type": "EvaluateFraudRisk",
    "@schemaLocation": "/schema-registry/EvaluateFraudRisk.yaml",
    "href": "/tmf-api/fraudManagement/v5/evaluateFraudRisk/f3a34f89-0046-tsk-01",
    "id": "f3a34f89-0046-tsk-01",
    "requestedFraudEvaluationDate": "2024-11-20T09:34:58Z",
    "state": "done",
    "fraudTargetIdentifier": [
      {
        "@type": "PhoneContactMedium",
        "@baseType": "ContactMedium",
        "@schemaLocation": "/schema-registry/PhoneContactMedium.yaml",
        "phoneNumber": "01 09 75 83 51"
      }
    ],
    "relatedParty": [
      {
        "@type": "RelatedPartyRefOrPartyRoleRef",
        "@schemaLocation": "/schema-registry/RelatedPartyRef.yaml",
        "partyOrPartyRole": {
          "@type": "PartyRef",
          "href": "/tmf-api/partyManagement/v5/Organization/f3a34f89-0046-pty-01",
          "id": "f3a34f89-0046-pty-01",
          "name": "Bank XYZ",
          "@referredType": "Organization"
        },
        "role": "Merchant"
      }
    ],
    "fraudEvaluationCriteria": {
      "@type": "FraudEvaluationCriteria",
      "@schemaLocation": "/schema-registry/FraudEvaluationCriteria.yaml",
      "fraudRiskProfileCriteria": {
        "@type": "FraudRiskProfileCriteria",
        "@schemaLocation": "/schema-registry/FraudRiskProfileCriteria.yaml",
        "simTenure": false,
        "deviceTenure": true,
        "lineTenure": false,
        "paymentMethod": false,
        "callForwarding": true,
        "serviceStatus": true,
        "simSwap": false
      },
      "fraudNetworkAuthenticationCriteria": {
        "@type": "FraudNetworkAuthenticationCriteria",
        "@schemaLocation": "/schema-registry/FraudNetworkAuthenticationCriteria.yaml",
        "callVerification": false
      },
```

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```
      "fraudScoreCriteria": {
        "@type": "FraudScoreCriteria",
        "@schemaLocation": "/schema-registry/FraudScoreCriteria.yaml",
        "isExplanationRequired": true
      },
      "fraudProfileMatchCriteria": {
        "@type": "FraudProfileMatchCriteria",
        "@schemaLocation": "/schema-registry/FraudProfileMatchCriteria.yaml",
        "email": {
          "@type": "EmailContactMedium",
          "@baseType": "ContactMedium",
          "@schemaLocation": "/schema-registry/EmailContactMedium.yaml",
          "emailAddress": "neo@matrix.com"
        },
        "partyName": "Joe Doe",
        "address": {
          "@type": "GeographicAddressContactMedium",
          "@baseType": "ContactMedium",
          "@schemaLocation": "/schema-registry/GeographicAddressContactMedium.yaml",
          "id": "address-001",
          "preferred": true,
          "contactType": "private",
          "validFor": {
            "startDateTime": "2025-01-29T10:40:52Z",
            "endDateTime": "2025-07-29T10:40:52Z"
          },
          "city": "Gurgaon",
          "country": "India",
          "postCode": "121001",
          "stateOrProvince": "Harayana",
          "street1": "Sector 39",
          "street2": "Near Unitech cyber park",
          "geographicAddress": {
            "@type": "GeographicAddressRef",
            "@schemaLocation": "/schema-registry/GeographicAddressRef.yaml",
            "href": "/tmf-
api/geographicAddressManagement/v5/geographicAddress/35dcfeec-9051-4b05-
830e7a0f67dc541d",
            "id": "35dcfeec-9051-4b05-830e7a0f67dc541d",
            "name": "address1",
            "@referredType": "GeographicAddress"
          }
        }
      }
    },
    "fraudEvaluationResult": {
      "@type": "FraudEvaluationResult",
      "@schemaLocation": "/schema-registry/FraudEvaluationResult.yaml",
      "fraudRiskProfileResult": {
        "@type": "FraudRiskProfileResult",
        "@schemaLocation": "/schema-registry/FraudRiskProfileResult.yaml",
        "simTenure": {
          "units": "month",
          "amount": 120
        },
        "deviceTenure": {
          "units": "month",
          "amount": 48
        },
        "lineTenure": {
```


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```
          "units": "month",
          "amount": 120
        },
        "paymentMethod": {
          "@type": "PaymentMethodRef",
          "id": "pm-cc-01",
          "name": "Credit Card",
          "@referredType": "BankCard"
        },
        "callForwardingStatus": true,
        "serviceStatus": "active",
        "role": "Admin",
        "simSwapStatus": false
      },
      "fraudNetworkAuthenticationResult": {
        "@type": "FraudNetworkAuthenticationResult",
        "@schemaLocation": "/schema-registry/FraudNetworkAuthenticationResult.yaml",
        "callVerification": "VERIFIED"
      },
      "fraudScoreResult": {
        "@type": "FraudScoreResult",
        "@schemaLocation": "/schema-registry/FraudScoreResult.yaml",
        "score": "70",
        "explanation": "Credit Score low"
      },
      "fraudProfileMatchResult": {
        "@type": "FraudProfileMatchResult",
        "globalMatchScore": "90",
        "profileMatch": [
          {
            "@type": "ProfileMatch",
            "matchElement": "name",
            "matchScore": "90"
          }
        ]
      }
    }
  }
]
```


Creates a evaluateFraudRisk

`POST /evaluateFraudRisk?fields=…&{filtering}`

Description

This operation creates an EvaluateFraudRisk entity.

Mandatory Attributes

Mandatory Attributes	Rule
@type	

Usage samples

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Submit a fraud risk evaluation request and receive immediate results, persisting the task for auditing purposes.

Request

```
POST /evaluateFraudRisk
Content-Type: application/json

{
  "@type": "EvaluateFraudRisk",
  "@schemaLocation": "/schema-registry/EvaluateFraudRisk.yaml",
  "requestedFraudEvaluationDate": "2024-11-20T09:34:58Z",
  "fraudTargetIdentifier": [
    {
      "@type": "PhoneContactMedium",
      "@baseType": "ContactMedium",
      "@schemaLocation": "/schema-registry/PhoneContactMedium.yaml",
      "phoneNumber": "01 09 75 83 51"
    }
  ],
  "relatedParty": [
    {
      "@type": "RelatedPartyRefOrPartyRoleRef",
      "@schemaLocation": "/schema-registry/RelatedPartyRef.yaml",
      "partyOrPartyRole": {
        "@type": "PartyRef",
        "href": "/tmf-api/partyManagement/v5/Organization/f3a34f89-0046-pty-01",
        "id": "f3a34f89-0046-pty-01",
        "name": "Bank XYZ",
        "@referredType": "Organization"
      },
      "role": "Merchant"
    }
  ],
  "fraudEvaluationCriteria": {
    "@type": "FraudEvaluationCriteria",
    "@schemaLocation": "/schema-registry/FraudEvaluationCriteria.yaml",
    "fraudRiskProfileCriteria": {
      "@type": "FraudRiskProfileCriteria",
      "@schemaLocation": "/schema-registry/FraudRiskProfileCriteria.yaml",
      "simTenure": false,
      "deviceTenure": true,
      "lineTenure": false,
      "paymentMethod": false,
      "callForwarding": true,
      "serviceStatus": true,
      "simSwap": false
    },
    "fraudNetworkAuthenticationCriteria": {
      "@type": "FraudNetworkAuthenticationCriteria",
      "@schemaLocation": "/schema-registry/FraudNetworkAuthenticationCriteria.yaml",
      "callVerification": false
    },
    "fraudScoreCriteria": {
      "@type": "FraudScoreCriteria",
      "@schemaLocation": "/schema-registry/FraudScoreCriteria.yaml",
      "isExplanationRequired": true
```

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```
    },
    "fraudProfileMatchCriteria": {
      "@type": "FraudProfileMatchCriteria",
      "@schemaLocation": "/schema-registry/FraudProfileMatchCriteria.yaml",
      "email": {
        "@type": "EmailContactMedium",
        "@baseType": "ContactMedium",
        "@schemaLocation": "/schema-registry/EmailContactMedium.yaml",
        "emailAddress": "neo@matrix.com"
      },
      "partyName": "Joe Doe",
      "address": {
        "@type": "GeographicAddressContactMedium",
        "@baseType": "ContactMedium",
        "@schemaLocation": "/schema-registry/GeographicAddressContactMedium.yaml",
        "id": "address-001",
        "preferred": true,
        "contactType": "private",
        "validFor": {
          "startDateTime": "2025-01-29T10:40:52Z",
          "endDateTime": "2025-07-29T10:40:52Z"
        },
        "city": "Gurgaon",
        "country": "India",
        "postCode": "121001",
        "stateOrProvince": "Harayana",
        "street1": "Sector 39",
        "street2": "Near Unitech cyber park",
        "geographicAddress": {
          "@type": "GeographicAddressRef",
          "@schemaLocation": "/schema-registry/GeographicAddressRef.yaml",
          "href": "/tmf-api/geographicAddressManagement/v5/geographicAddress/35dcfeec-
9051-4b05-830e7a0f67dc541d",
          "id": "35dcfeec-9051-4b05-830e7a0f67dc541d",
          "name": "address1",
          "@referredType": "GeographicAddress"
        }
      }
    }
  }
}
```


Response


```
201

{
  "@type": "EvaluateFraudRisk",
  "@schemaLocation": "/schema-registry/EvaluateFraudRisk.yaml",
  "href": "/tmf-api/fraudManagement/v5/evaluateFraudRisk/f3a34f89-0046-tsk-01",
  "id": "f3a34f89-0046-tsk-01",
  "requestedFraudEvaluationDate": "2024-11-20T09:34:58Z",
  "state": "done",
  "fraudTargetIdentifier": [
    {
      "@type": "PhoneContactMedium",
      "@baseType": "ContactMedium",
```


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```
      "@schemaLocation": "/schema-registry/PhoneContactMedium.yaml",
      "phoneNumber": "01 09 75 83 51"
    }
  ],
  "relatedParty": [
    {
      "@type": "RelatedPartyRefOrPartyRoleRef",
      "@schemaLocation": "/schema-registry/RelatedPartyRef.yaml",
      "partyOrPartyRole": {
        "@type": "PartyRef",
        "href": "/tmf-api/partyManagement/v5/Organization/f3a34f89-0046-pty-01",
        "id": "f3a34f89-0046-pty-01",
        "name": "Bank XYZ",
        "@referredType": "Organization"
      },
      "role": "Merchant"
    }
  ],
  "fraudEvaluationCriteria": {
    "@type": "FraudEvaluationCriteria",
    "@schemaLocation": "/schema-registry/FraudEvaluationCriteria.yaml",
    "fraudRiskProfileCriteria": {
      "@type": "FraudRiskProfileCriteria",
      "@schemaLocation": "/schema-registry/FraudRiskProfileCriteria.yaml",
      "simTenure": false,
      "deviceTenure": true,
      "lineTenure": false,
      "paymentMethod": false,
      "callForwarding": true,
      "serviceStatus": true,
      "simSwap": false
    },
    "fraudNetworkAuthenticationCriteria": {
      "@type": "FraudNetworkAuthenticationCriteria",
      "@schemaLocation": "/schema-registry/FraudNetworkAuthenticationCriteria.yaml",
      "callVerification": false
    },
    "fraudScoreCriteria": {
      "@type": "FraudScoreCriteria",
      "@schemaLocation": "/schema-registry/FraudScoreCriteria.yaml",
      "isExplanationRequired": true
    },
    "fraudProfileMatchCriteria": {
      "@type": "FraudProfileMatchCriteria",
      "@schemaLocation": "/schema-registry/FraudProfileMatchCriteria.yaml",
      "email": {
        "@type": "EmailContactMedium",
        "@baseType": "ContactMedium",
        "@schemaLocation": "/schema-registry/EmailContactMedium.yaml",
        "emailAddress": "neo@matrix.com"
      },
      "partyName": "Joe Doe",
      "address": {
        "@type": "GeographicAddressContactMedium",
        "@baseType": "ContactMedium",
        "@schemaLocation": "/schema-registry/GeographicAddressContactMedium.yaml",
        "id": "address-001",
        "preferred": true,
        "contactType": "private",
        "validFor": {
```


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```
          "startDateTime": "2025-01-29T10:40:52Z",
          "endDateTime": "2025-07-29T10:40:52Z"
        },
        "city": "Gurgaon",
        "country": "India",
        "postCode": "121001",
        "stateOrProvince": "Harayana",
        "street1": "Sector 39",
        "street2": "Near Unitech cyber park",
        "geographicAddress": {
          "@type": "GeographicAddressRef",
          "@schemaLocation": "/schema-registry/GeographicAddressRef.yaml",
          "href": "/tmf-api/geographicAddressManagement/v5/geographicAddress/35dcfeec-
9051-4b05-830e7a0f67dc541d",
          "id": "35dcfeec-9051-4b05-830e7a0f67dc541d",
          "name": "address1",
          "@referredType": "GeographicAddress"
        }
      }
    }
  },
  "fraudEvaluationResult": {
    "@type": "FraudEvaluationResult",
    "@schemaLocation": "/schema-registry/FraudEvaluationResult.yaml",
    "fraudRiskProfileResult": {
      "@type": "FraudRiskProfileResult",
      "@schemaLocation": "/schema-registry/FraudRiskProfileResult.yaml",
      "simTenure": {
        "units": "month",
        "amount": 120
      },
      "deviceTenure": {
        "units": "month",
        "amount": 48
      },
      "lineTenure": {
        "units": "month",
        "amount": 120
      },
      "paymentMethod": {
        "@type": "PaymentMethodRef",
        "id": "pm-cc-01",
        "name": "Credit Card",
        "@referredType": "BankCard"
      },
      "callForwardingStatus": true,
      "serviceStatus": "active",
      "role": "Admin",
      "simSwapStatus": false
    },
    "fraudNetworkAuthenticationResult": {
      "@type": "FraudNetworkAuthenticationResult",
      "@schemaLocation": "/schema-registry/FraudNetworkAuthenticationResult.yaml",
      "callVerification": "VERIFIED"
    },
    "fraudScoreResult": {
      "@type": "FraudScoreResult",
      "@schemaLocation": "/schema-registry/FraudScoreResult.yaml",
      "score": "70",
      "explanation": "Credit Score low"
```


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```
    },
    "fraudProfileMatchResult": {
      "@type": "FraudProfileMatchResult",
      "globalMatchScore": "90",
      "profileMatch": [
        {
          "@type": "ProfileMatch",
          "matchElement": "name",
          "matchScore": "90"
        }
      ]
    }
  }
}
```


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API NOTIFICATIONS

For every single of operation on the entities use the following templates and provide sample REST notification POST calls.

It is assumed that the Pub/Sub uses the Register and UnRegister mechanisms described in the REST Guidelines reproduced below.

Register listener

POST /hub

Description

Sets the communication endpoint address the service instance must use to deliver information about its health state, execution state, failures and metrics. Subsequent POST calls will be rejected by the service if it does not support multiple listeners. In this case DELETE /api/hub/{id} must be called before an endpoint can be created again.

Behavior

Returns HTTP/1.1 status code 204 if the request was successful.

Returns HTTP/1.1 status code 409 if request is not successful.

Usage Samples

Here's an example of a request for registering a listener.

Request

```
POST /api/hub

Accept: application/json

{
    "callback": "http://in.listener.com"
}
```

Response

```
201

Content-Type: application/json

Location: /api/hub/42
```

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```
{
    "id":"42",
    "callback": "http://in.listener.com",
    "query":""
}
```

Unregister listener

DELETE /hub/{id}

Description

Clears the communication endpoint address that was set by creating the Hub..

Behavior

Returns HTTP/1.1 status code 204 if the request was successful.

Returns HTTP/1.1 status code 404 if the resource is not found.

Usage Samples

Here's an example of a request for un-registering a listener.

Request

```
DELETE /api/hub/42

Accept: application/json
```

Response

```
204
```

Publish Event to listener

POST /client/listener

Description

Clears the communication endpoint address that was set by creating the Hub.

Provides to a registered listener the description of the event that was raised. The /client/listener url is the callback url passed when registering the listener.

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Behavior

Returns HTTP/1.1 status code 201 if the service is able to set the configuration.

Usage Samples

Here's an example of a notification received by the listener. In this example "EVENT TYPE" should be replaced by one of the notification types supported by this API (see Notification resources Models section) and EVENT BODY refers to the data structure of the given notification type.

Request

```
POST /client/listener

Accept: application/json

{
    "event": {
        EVENT BODY
    },
    "eventType": "EVENT_TYPE"
}
```

Response

```
201
```

For detailed examples on the general TM Forum notification mechanism, see the TMF REST Design Guidelines.

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Acknowledgements

Release History

Release Number	Date	Release led by:	Description
Release 5.0.0	08/06/2026	Subhanshu Shukla Jio Platforms Ltd.	• Initial release of the user guide for TMF770 Fraud Management API • Alignment with OpenAPI v5 guidelines

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